Enterprise Network Lab

CCNA Practice & Portfolio Project

Three-Tier Hierarchical Design | Cisco Packet Tracer

1. Project Overview

This lab simulates a production-grade enterprise network using Cisco Packet Tracer. It is designed to cover the majority of CCNA exam topics in a single cohesive topology, and to serve as a portfolio piece demonstrating real-world networking knowledge.

Technologies implemented:

- VLANs and trunking (802.1Q)
- Rapid Per-VLAN Spanning Tree (RPVST+)
- Inter-VLAN routing via SVIs on Layer 3 switches
- OSPF dynamic routing
- DHCP — one pool per VLAN
- FHRP (HSRP) — redundant default gateways
- NAT/PAT — internet access via R3
- ACLs — standard and extended
- Port security — sticky MAC addresses
- SSH — hardened remote management

2. Topology

2.1 Device Summary

Layer	Devices	Role	Notes
Core	R3	Edge / Internet Gateway	Connects to ISP, NAT/PAT
Distribution	R1, R2	WAN / OSPF backbone	Route between DS switches
Distribution	DS1, DS2 (3560)	Inter-VLAN routing, FHRP, STP root	Layer 3 switches — SVIs
Access	AS1–AS8	Host connectivity	RPVST+, port security, VLANs

2.2 Device Roles

- R3 — Edge router, connects to ISP. Handles NAT/PAT overload and OSPF redistribution.
- R1 / R2 — Distribution routers, OSPF peers with R3 and DS switches.
- DS1 / DS2 — Cisco 3560 Layer 3 switches. Run SVIs for inter-VLAN routing, HSRP for gateway redundancy, and act as STP root per VLAN.
- AS1–AS8 — Access layer switches. Connect end hosts, run RPVST+, enforce port security.

3. VLAN Design

VLAN ID	Name	Department	Hosts	Color Code
VLAN 10	IT	IT Dept	105	Blue
VLAN 20	SALES	Sales Dept	47	Purple
VLAN 30	HR	HR Dept	20	Red
VLAN 40	MGMT	Management	10	Green

Note: VLANs span both sides of the topology. DS1 serves the left side (AS1–AS4), DS2 serves the right side (AS5–AS8). Inter-VLAN traffic between sides is routed via OSPF through R1 and R2.

4. IP Address Plan

4.1 Address Space

Base: 10.0.0.0/8 — VLSM applied to size subnets per department.

4.2 Subnet Table

Segment	Subnet	Network	First Usable	Last Usable	Hosts
IT (VLAN 10)	/25	10.0.0.0	10.0.0.1	10.0.0.126	126
Sales (VLAN 20)	/26	10.0.0.128	10.0.0.129	10.0.0.190	62
HR (VLAN 30)	/27	10.0.0.192	10.0.0.193	10.0.0.222	30
MGMT (VLAN 40)	/28	10.0.0.224	10.0.0.225	10.0.0.238	14
R1 — R2	/30	10.0.0.240	10.0.0.241	10.0.0.242	2
R1 — R3	/30	10.0.0.244	10.0.0.245	10.0.0.246	2
R2 — R3	/30	10.0.0.248	10.0.0.249	10.0.0.250	2
R3 — ISP	/30	10.0.0.252	10.0.0.253	10.0.0.254	2

DS1 — R1	/30	10.0.1.0	10.0.1.1	10.0.1.2	2
DS2 — R2	/30	10.0.1.4	10.0.1.5	10.0.1.6	2

4.3 SVI Gateway Addresses

DS1 holds the HSRP Active gateway, DS2 holds Standby. Convention: first usable = DS1 SVI, second usable = DS2 SVI, virtual IP = third usable.

VLAN	DS1 SVI	DS2 SVI	HSRP VIP	Subnet
VLAN 10 — IT	10.0.0.1	10.0.0.2	10.0.0.3	10.0.0.0/25
VLAN 20 — Sales	10.0.0.129	10.0.0.130	10.0.0.131	10.0.0.128/26
VLAN 30 — HR	10.0.0.193	10.0.0.194	10.0.0.195	10.0.0.192/27
VLAN 40 — MGMT	10.0.0.225	10.0.0.226	10.0.0.227	10.0.0.224/28

5. Configuration Stages

#	Stage	Description	Status
1	Basic Device Setup	Hostnames, passwords, SSH, disable Telnet, MOTD banner, save config on all devices	**Complete**
2	VLANs	Create VLANs on all switches, assign access ports, configure trunk ports	**Complete**
3	Inter-VLAN Routing	SVIs on DS1 and DS2 (multilayer switches), enable ip routing	**Complete**
4	STP / RPVST+	Configure RPVST+, set STP root per VLAN, align with FHRP active gateway	**Complete**
5	FHRP (HSRP)	DS1/DS2 as redundant gateways per VLAN, load balance across VLANs	**Complete**
6	OSPF	R1, R2, R3, DS1, DS2 — single area or multi-area OSPF	**Complete**
7	DHCP	One pool per VLAN, configure helpers if centralised	**Complete**
8	NAT/PAT	R3 to ISP — PAT overload for internet access	**Complete**
9	ACLs	Standard and extended — block inter-dept traffic where required	**PT Limitation**
10	Port Security	Sticky MAC on all access ports	**Complete**
11	SSH Hardening	Disable Telnet, enforce SSHv2, set domain name, generate keys	**Complete**
12	Documentation	GitHub README, topology diagram, config files	**In Progress**

6. Design Decisions & Notes

STP + FHRP Alignment

STP root bridge and HSRP active gateway must be the same device per VLAN to prevent suboptimal traffic paths. DS1 is root and active for VLANs 10 and 30. DS2 is root and active for VLANs 20 and 40. This achieves load balancing across both distribution switches.

EtherChannel

Attempted LACP EtherChannel between DS1 (3560) and AS2 (2960). Persistent SD/suspended state due to Packet Tracer compatibility issues between mixed switch models (3560 and 2960). Feature scrapped — not worth forcing in a mixed-model environment. In a real network, EtherChannel would be configured between same-model switches at distribution layer.

NAT/PAT

R3 performs PAT overload on its WAN interface toward ISP. All internal VLANs are translated. OSPF default route is redistributed from R3 into the internal routing domain.

OSPF Design

Single area (Area 0) across R1, R2, R3, DS1, DS2. DS switches participate in OSPF to advertise VLAN subnets. R3 redistributes the default route.

7. Progress Tracker

- Planning & Topology Design — COMPLETE
- VLAN Design — COMPLETE
- IP Addressing / Subnetting — COMPLETE
- Basic Device Setup — COMPLETE
- VLAN Configuration — COMPLETE
- EtherChannel — SCRAPPED (PT 3560/2960 compatibility issue)
- Inter-VLAN Routing (SVIs) — COMPLETE
- STP / RPVST+ — COMPLETE
- FHRP (HSRP) — COMPLETE (VLAN 20 PT bug documented)
- OSPF — COMPLETE
- DHCP — COMPLETE
- NAT/PAT — COMPLETE
- ACLs — ATTEMPTED / PT LIMITATION (full documentation in Section 15)
- Port Security — COMPLETE
- Documentation & GitHub — IN PROGRESS

8. Stage 1 — Basic Device Configuration

Applied to: R1, R2, R3, DS1, DS2, AS1-AS8. See variations per device type below.

8.1 Router Base Config (R1, R2, R3)

Commands executed in order on each router:

Command	Purpose
en	Enter privileged EXEC mode
conf t	Enter global configuration mode
hostname <R1 / R2 / R3>	Set device hostname
enable secret CCNA	Set encrypted privileged password
username admin secret CCNA	Create local user for SSH/console auth
no ip domain-lookup	Prevent router hanging on mistyped commands
ip domain-name lab.local	Required for RSA key generation
banner motd #Unauthorized access prohibited. #	Legal warning banner
line con 0	Enter console line config
login local	Require local username/password on console
exec-timeout 5 0	Auto-logout after 5 minutes idle
logging synchronous	Prevent log messages interrupting typing
exit	Return to global config
line vty 0 15	Enter all virtual terminal lines
login local	Require local auth on VTY
exec-timeout 5 0	Auto-logout after 5 minutes idle
transport input ssh	Disable Telnet - SSH only
exit	Return to global config
crypto key generate rsa (2048 bits)	Generate RSA keypair for SSH
ip ssh version 2	Force SSHv2 only
ip ssh time-out 60	SSH session timeout
ip ssh authentication-retries 3	Limit brute force attempts
do wr	Save config to NVRAM

8.2 Variations per Device Type

- DS1, DS2 (Layer 3 switches) - add 'ip routing' after hostname config.
- AS1-AS8 (Access switches) - same base config, no 'ip routing', no routed interfaces.
- ISP (PT Server) - static IP set via Desktop > IP Configuration. No CLI config required.

8.3 Router Interface IP Addresses

Device	Interface	IP Address	Connects To
R1	s0/1/0	10.0.0.245 /30	R3
R1	s0/1/1	10.0.0.241 /30	R2
R1	g0/0/0	10.0.1.1 /30	DS1
R2	s0/1/0	10.0.0.249 /30	R3
R2	s0/1/1	10.0.0.242 /30	R1
R2	g0/0/0	10.0.1.5 /30	DS2
R3	s0/1/0	10.0.0.246 /30	R1
R3	s0/1/1	10.0.0.250 /30	R2
R3	g0/0	10.0.0.253 /30	ISP
ISP	NIC	10.0.0.254 /30	R3 g0/0

9. Stage 2 — VLAN & Port Configuration

9.1 VLANs Created on All Switches (DS1, DS2, AS1-AS8)

VLAN ID	Name	Purpose
10	IT	IT department data traffic
20	SALES	Sales department data traffic
30	HR	HR department data traffic
40	MGMT	Management department data traffic
50	VOICE	VoIP traffic for IP phones
99	NATIVE	Native VLAN for trunk ports (security hardening)

9.2 Access Switch Port Assignments

Switch	Interface(s)	Mode	VLAN	Notes

AS1	Fa0/1-2	Access	VLAN 10 - IT	PC-IT-1, PC-IT-2
AS1	Fa0/3-5	Trunk	10,20,30,40,50,99	Native VLAN 99
AS2	Fa0/1	Access	VLAN 40 - MGMT	PC-MGMT-1
AS2	Fa0/2-6	Trunk	10,20,30,40,50,99	Native VLAN 99
AS3	Fa0/1	Access	VLAN 30 - HR	PC-HR-1
AS3	Fa0/2	Access	VLAN 30 + VOICE 50	Phone-HR-1
AS3	Fa0/3-5	Trunk	10,20,30,40,50,99	Native VLAN 99
AS4	Fa0/1	Access	VLAN 20 - Sales	PC-Sales-1
AS4	Fa0/2	Access	VLAN 20 + VOICE 50	Phone-Sales-1
AS4	Fa0/3-5	Trunk	10,20,30,40,50,99	Native VLAN 99
AS5	Fa0/1-2	Access	VLAN 10 - IT	PC-IT-3, PC-IT-4
AS5	Fa0/3-5	Trunk	10,20,30,40,50,99	Native VLAN 99
AS6	Fa0/1	Access	VLAN 40 - MGMT	PC-MGMT-2
AS6	Fa0/2-6	Trunk	10,20,30,40,50,99	Native VLAN 99
AS7	Fa0/1	Access	VLAN 20 - Sales	PC-Sales-3
AS7	Fa0/2	Access	VLAN 20 + VOICE 50	Phone-Sales-2
AS7	Fa0/3-5	Trunk	10,20,30,40,50,99	Native VLAN 99
AS8	Fa0/1	Access	VLAN 30 - HR	PC-HR-2
AS8	Fa0/2	Access	VLAN 30 + VOICE 50	Phone-HR-2
AS8	Fa0/3-5	Trunk	10,20,30,40,50,99	Native VLAN 99

9.3 Security and STP Features Applied

- PortFast - enabled on all access ports (immediate forwarding, no listening/learning delay)
- BPDU Guard - enabled on all access ports (err-disable port if BPDU received from rogue switch)
- DTP Disabled - switchport nonegotiate on all trunk ports (prevents dynamic trunk negotiation attacks)
- Native VLAN 99 - all trunk ports use VLAN 99 as native (untagged traffic isolated from data VLANs)
- dot1q Encapsulation - explicitly set on DS1 and DS2 trunk interfaces (required on L3 switches)

9.4 Distribution Switch Port Mappings

Switch	Interface(s)	Connects To	Mode
DS1	Fa0/1-2	AS2	Trunk, dot1q, Native 99, nonegotiate
DS1	Fa0/4-5	AS4	Trunk, dot1q, Native 99, nonegotiate
DS2	Fa0/2-3	AS6	Trunk, dot1q, Native 99, nonegotiate

DS2	Fa0/4-5	AS7	Trunk, dot1q, Native 99, nonegotiate

10. Stage 3 — SVIs & Stage 4 — RPVST+

10.1 SVIs on DS1 and DS2

SVIs provide Layer 3 default gateways for each VLAN. DS1 takes first usable IP per subnet, DS2 takes second usable. ip routing enabled on both switches.

VLAN	DS1 SVI	DS2 SVI	Subnet Mask
VLAN 10 - IT	10.0.0.1	10.0.0.2	255.255.255.128 /25
VLAN 20 - Sales	10.0.0.129	10.0.0.130	255.255.255.192 /26
VLAN 30 - HR	10.0.0.193	10.0.0.194	255.255.255.224 /27
VLAN 40 - MGMT	10.0.0.225	10.0.0.226	255.255.255.240 /28

10.2 RPVST+ Configuration

Rapid Per-VLAN Spanning Tree Plus configured on DS1 and DS2. Load balanced across both distribution switches per VLAN.

VLAN	Primary Root	Secondary Root	Priority
VLAN 10 - IT	DS1	DS2	24576 / 28672
VLAN 20 - Sales	DS2	DS1	24576 / 28672
VLAN 30 - HR	DS1	DS2	24576 / 28672
VLAN 40 - MGMT	DS2	DS1	24576 / 28672

10.3 Topology Changes for RPVST+

- Added cross-links: AS4 to DS2 and AS7 to DS1 — creates L2 redundant paths between both distribution switches
- Removed direct DS1-DS2 link — traffic now flows through access layer cross-links
- All new cross-links configured as trunks with dot1q, native VLAN 99, allowed VLANs, nonegotiate
- Root Guard removed from DS downlink ports — required because DS2 BPDUs need to reach DS1 through cross-links and vice versa

10.4 Lessons Learned

- RPVST+ requires active L2 paths between switches to exchange BPDUs — SVIs alone are not enough
- Root Guard must not be placed on inter-distribution links — only on pure access-facing ports
- STP root bridge and traffic flow must be deliberately designed, not left to default election

11. Stage 5 — FHRP (HSRP)

11.1 HSRP Design

HSRP configured on DS1 and DS2 SVIs providing redundant default gateways per VLAN. Active/Standby roles aligned with STP root bridge assignment for optimal traffic flow.

VLAN	Virtual IP	DS1 Role	DS2 Role	Priority
VLAN 10 - IT	10.0.0.3	Active	Standby	DS1=110, DS2=90
VLAN 20 - Sales	10.0.0.131	Standby	Active	DS1=90, DS2=110
VLAN 30 - HR	10.0.0.195	Active	Standby	DS1=110, DS2=90
VLAN 40 - MGMT	10.0.0.227	Standby	Active	DS1=90, DS2=110

11.2 HSRP Configuration Notes

- Preempt enabled on all groups - active router reclaims role after recovery
- HSRP group number matches VLAN ID for clarity and consistency
- Virtual IPs are third usable address in each subnet - used as default gateway by all hosts
- HSRP aligned with STP - DS1 active for VLANs 10,30 (also STP root), DS2 active for VLANs 20,40 (also STP root)

11.3 Known Issues

- VLAN 20 HSRP shows split-brain in Packet Tracer - both DS1 and DS2 show Active state with unknown standby. Config is correct, issue is PT not propagating HSRP multicast hellos for VLAN 20. VLANs 10, 30, 40 operating correctly.

12. Stage 6 — OSPF

12.1 OSPF Design

Single area OSPF (Area 0) configured across all 5 routing devices. DS1 and DS2 advertise VLAN subnets. R3 redistributes default route toward ISP.

Device	Router ID	Networks Advertised
DS1	1.1.1.1	10.0.0.0/25, 10.0.0.128/26, 10.0.0.192/27, 10.0.0.224/28, 10.0.1.0/30
DS2	2.2.2.2	10.0.0.0/25, 10.0.0.128/26, 10.0.0.192/27, 10.0.0.224/28, 10.0.1.4/30
R1	3.3.3.3	10.0.0.240/30, 10.0.0.244/30, 10.0.1.0/30
R2	4.4.4.4	10.0.0.240/30, 10.0.0.248/30, 10.0.1.4/30
R3	5.5.5.5	10.0.0.244/30, 10.0.0.248/30 + default-information originate

12.2 OSPF Notes

- Process ID 1 on all devices — locally significant only
- Router IDs manually configured — prevents ID changes if interfaces go down
- R3 has static default route to ISP (10.0.0.254) and redistributes it via default-information originate
- All devices receive O*E2 default route pointing to R3 as gateway of last resort
- OSPF adjacencies forming over VLAN SVIs between DS1 and DS2 — functional but not ideal, result of no dedicated L3 link between distribution switches

13. Stage 7 — DHCP

13.1 DHCP Design

DHCP pools configured on both DS1 and DS2 for all VLANs including Voice. Identical pools on both switches provide redundancy. Hosts use HSRP virtual IPs as default gateways.

Pool	Network	Gateway	DNS	Excluded
VLAN_10	10.0.0.0/25	10.0.0.3	8.8.8.8	10.0.0.1-3
VLAN_20	10.0.0.128/26	10.0.0.131	8.8.8.8	10.0.0.129-131
VLAN_30	10.0.0.192/27	10.0.0.195	8.8.8.8	10.0.0.193-195
VLAN_40	10.0.0.224/28	10.0.0.227	8.8.8.8	10.0.0.225-227
VLAN_50	10.0.1.8/29	10.0.1.9	8.8.8.8	None

13.2 Verification

- PC-IT-1 received 10.0.0.4/25 via DHCP with correct gateway (10.0.0.3) and DNS (8.8.8.8)

- All PCs and IP phones set to DHCP and receiving addresses successfully
- End-to-end connectivity verified — PC-IT-1 can ping ISP (10.0.0.254)
- Inter-VLAN routing confirmed — PC-IT-1 successfully pings Sales VLAN hosts

14. Stage 8 — NAT/PAT

14.1 NAT/PAT Design

PAT overload configured on R3. All internal traffic destined for outside is translated to R3's single public-facing interface IP (10.0.0.253). Uses standard ACL to define which traffic gets translated.

Interface	Direction	IP	Role
S0/1/0	Inside	10.0.0.246	Facing R1
S0/1/1	Inside	10.0.0.250	Facing R2
G0/0/0	Outside	10.0.0.253	Facing ISP — PAT address

14.2 NAT Configuration

- ACL 1 permits 10.0.0.0 0.0.255.255 — matches all internal traffic
- ip nat inside source list 1 interface GigabitEthernet0/0/0 overload — PAT overload to outside interface
- No NAT pool required — PAT uses single outside interface IP with port multiplexing

14.3 Verification

- show ip nat translations confirms PC-IT-1 (10.0.0.4) translated to R3 outside IP (10.0.0.253)
- PAT port numbers visible in translation table confirming overload operation

15. Stage 9 — Access Control Lists (ACLs)

15.1 Security Policy Design

The following inter-VLAN security policy was designed and implemented:

- IT — can only communicate within IT VLAN
- Sales — can only communicate within Sales VLAN
- HR — can only communicate within HR VLAN
- MGMT — can reach all VLANs (privileged management access)

- No department can initiate connections to MGMT — one way access only
- Voice VLAN 50 — unrestricted, QoS prioritized over filtering

15.2 ACL Design Decisions

- Standard ACLs chosen over Extended — filtering is per-VLAN regardless of protocol, no need for port/protocol granularity
- Applied outbound on destination SVI — standard ACLs placed close to destination per best practice
- Two permit statements per ACL — own subnet + MGMT subnet, implicit deny handles rest
- Named ACLs attempted first for readability, then numbered ACLs as fallback

15.3 ACL Logic (Correct Design)

ACL	Applied To	Direction	Permits
ACL 10 / ACL_VLAN10	interface vlan 10	Outbound	10.0.0.0/25 (IT), 10.0.0.224/28 (MGMT)
ACL 20 / ACL_VLAN20	interface vlan 20	Outbound	10.0.0.128/26 (Sales), 10.0.0.224/28 (MGMT)
ACL 30 / ACL_VLAN30	interface vlan 30	Outbound	10.0.0.192/27 (HR), 10.0.0.224/28 (MGMT)
ACL 40 / ACL_VLAN40	interface vlan 40	Outbound	10.0.0.224/28 (MGMT only)
ACL 50 / ACL_VLAN50	interface vlan 50	Outbound	any (voice unrestricted)

15.4 Commands Attempted

Named ACL attempt (first approach):

- ip access-list standard ACL_VLAN10
- permit 10.0.0.0 0.0.0.127
- permit 10.0.0.224 0.0.0.15
- interface vlan 10 / ip access-group ACL_VLAN10 out

Numbered ACL attempt (fallback approach):

- access-list 10 permit 10.0.0.0 0.0.0.127
- access-list 10 permit 10.0.0.224 0.0.0.15
- interface vlan 10 / ip access-group 10 out
- Inbound direction also attempted as alternative — ip access-group 10 in

Troubleshooting steps taken:

- Verified ACLs created correctly with show ip access-lists — all permit statements present
- Verified interface with show ip interface vlan 10 — outgoing access list showed as not set despite applying

- Verified running-config interface vlan 10 — ip access-group line absent from config
- Tried named ACLs, numbered ACLs, inbound and outbound directions — all failed to stick

15.5 Packet Tracer Limitation

After extensive troubleshooting, confirmed that Cisco Packet Tracer does not support ACL application on SVI (Switched Virtual Interface) interfaces on the 3560 multilayer switch model. The ACL commands are accepted without error but silently discarded — they do not appear in running-config and have no effect on traffic.

This is a known PT simulator limitation. In a real Cisco 3560 or equivalent multilayer switch, the exact same configuration would work correctly. The ACL design, policy logic, wildcard masks, and application direction are all correct and verified.

In a production environment, this security policy would be implemented using one of the following alternatives:

- ACLs on SVI interfaces on real hardware — identical commands, fully supported
- Router-on-a-stick with subinterfaces — ACLs applied per subinterface
- VLAN ACLs (VACLs) using vlan access-map — switches traffic within a VLAN
- Private VLANs (PVLANs) — hardware-level isolation between hosts

16. Stage 10 — Port Security: Verification

- show port-security confirms enabled on all access ports with violation mode shutdown
- Sticky MACs learned after traffic generated — CurrentAddr count increments per active port
- Trunk ports correctly excluded — applying to trunks caused immediate err-disable due to multiple MACs